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How did you like using a screen reader?

I attempted to traverse the University of South Carolina website (www.sc.edu) using VoiceOver and NVDA screen reader applications. The amount of information presented made traversal and comprehension of the site mentally and cognitively taxing. For example, the amount of metadata transferred from code to audio can be overwhelming and excruciating. Furthermore, my anxiety and frustration grew the more I tried, unsuccessfully, to navigate web pages and interact with content. For example, on more than one occasion, I'd see interesting news or press releases, but I could not navigate the different sections of the content because the paragraphs and media were not annotated. As a novice screen reader, I was limited in my ability to quickly assert my location on screen or leverage many keyboard shortcuts to move between content sections. While the site was mostly consistent by using headers and footers, the main section of the page was not always segmented well.

What are the challenges of using a screen reader?

The challenges arise from attempting to render, interpret, or otherwise translate an infinite number of combinations of HTML code and attributes into intelligible chunks of information. Currently, standards such as the W3C's Web Accessibility Initiative (WAI) and Web Content Accessibility Guidelines (WCAG) are helpful in describing programmatic and design "best practices", but are limited by dependencies on other WAI guidelines, ambiguity, complexity, logical flaws, and guideline comprehension (Sloan et al., 2006). Furthermore, the amount of cognitive and mental effort to use screen readers is astronomical, not to mention that even proficient screen reader users lose a significant amount of time due to confusing layouts, poor labeling, and/or unhandled screen reader issues (Kearney-Volpe & Hurst, 2021). Unfortunately, web developers are not instructed to prioritize accessibility features over functional issues, which can result in flawed HTML in which screen reader metadata doesn't exist because the information was not coded. Flawed code can also lead to screen reader users often blaming themselves for not finding or accessing information (Bigham et al., 2017). While there have been attempts to extend and enhance screen reader technology by using browser extensions and other methods including semantic content annotations to provide meaningful segments of information, the challenge to capture and adapt to a combination of user behavior and dynamic page layout is confounding and ongoing (Lee & Ashok, 2020).

Do you have any recommendations for web developers, teachers or instructional designers after using the screen reader to access information?

For developers, while code and content should conform to WAI and WCAG standards, product and business owners can prioritize accessibility and screen reader compatibility. Teachers and instructional designers that write content and develop web applications should also consider screen reader compatibility as an essential component of a sustainable differentiated learning environment. Accessibility-first design for web developers and instructional staff overlaps usability, and allowing opportunities to experience their applications using one or more screen readers provides insights that otherwise are ignored (Youngblood, 2013). Explicit and intentional use of semantic structure, alternative content (e.g., captions, icons, images), keyboard (e.g., tab order) navigation, and routine accessibility maintenance as a function of the application lifecycle can reduce time-based omissions and make a significant impact in the effectiveness of screen readers and other accessibility tools (Trewin et al., 2010; Ferdous et al., 2021). In addition, all stakeholders can agree to prioritizing users and assistive technologies as a primary, versus secondary or tertiary, part of application development (Petrie et al., 2015).

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